

Stock-Market Dynamics: A Graph-Based Cluster-Dynamics Model of S&P 500 Market Regimes, 2000–2021

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Technical documentation, September 2026

Adapted from the graph-based cluster-dynamics framework (RadCluster): the market-regime bin is a *cluster*, the stocks are the migrating particles, and the “reactions” are regime drift, herding flows, and index entry/exit.

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1 Introduction

This document describes a stock-market prototype built by reusing the graph-based modeling approach of RadCluster: the market is partitioned into discrete regime classes (vertices), and every capital-migration process is an admissible transition (edge) carrying a rate kernel. The state is integrated as a system of ordinary differential equations over 2000–2021, and validated against two hindcast episodes: the 2020–21 COVID concentration and the 2008–09 financial-crisis drawdown.

In the language of the cluster-dynamics framework, the market-regime bin is a *cluster*: each stock belongs to a cluster R_{mv} labeled by its momentum bin $m = 0, \dots, 4$ and volatility bin $v = 0, 1, 2$, the direct analog of the cluster size in the materials formulation. The stocks are the migrating particles; $\mathbf{c}(t) \in \mathbb{R}^{15}$ is the distribution of index stocks across the 15 regime clusters. The bins use *absolute* thresholds, not cross-sectional ranks: momentum is the trailing 12-minus-1 month return on adjusted close in the bands $< 0\%$, $0\text{--}15\%$, $15\text{--}30\%$, $30\text{--}60\%$, $> 60\%$; volatility is the trailing-60-trading-day annualized standard deviation of daily log returns in the terciles $< 20\%$, $20\text{--}32\%$, $> 32\%$. Absolute bins are essential: quintile ranks would freeze every marginal at 20% by construction and the distribution could never concentrate, which is the phenomenon under study.

Three edge classes act on the graph:

- **Drift** — the measured month-to-month Markov transition matrix $\mathbf{T}$: momentum builds or decays and volatility shifts, so stocks wander between adjacent regime bins. A single stationary $\mathbf{T}$ is estimated on 2000–2019, and a regime-dependent pair $(\mathbf{T}_{\text{calm}}, \mathbf{T}_{\text{stress}})$ split by the VIX level is carried as the extension.
- **Herding** — the bilinear winner-chasing flux at rate h , the market analog of the assortativity/mating term: capital flows toward recent winners, the candidate mechanism for concentration episodes. $h = 0$ is the drift-only null.
- **Source/sink** — index additions (IPOs, promotions) and removals (delistings, acquisitions), measured but tiny.

The forecasting objective is the *future regime distribution*: what fraction of the market sits in each bin next month, and in particular its *concentration* — whether capital crowds into winner bins or disperses. Individual stock prices are not predicted.

Two hindcast cases are carried: **Case 1**, the 2020–21 COVID episode (a recovery-driven concentration: 55% of stocks above 60% twelve-month return in March 2021, measured off the crash bottom); and **Case 2**, the 2008–09 financial crisis (a drawdown: 95% of stocks in the loser bin in late 2008). Both are run out-of-sample from pre-episode calibration.

2 Related work

The momentum leg of the model rests on the cross-sectional momentum literature: Jegadeesh and Titman [1] documented that buying recent winners and selling recent losers earns abnormal returns over 3–12 month horizons, the empirical basis for treating trailing return as a regime coordinate. Fama and French [2] systematized common risk factors in equity returns; volatility as the second regime axis follows the same risk-factor logic.

The herding term belongs to the econophysics tradition of imitation-driven market models. Cont and Bouchaud [3] showed that random percolation-like herding generates the heavy-tailed return distributions seen in real markets from purely imitative agents. Lux and Marchesi [4] built a

stochastic multi-agent model in which chartist–fundamentalist switching produces scaling and volatility clustering; the herding flux here is the mean-field, cluster-level counterpart of their switching dynamics. At the behavioral level, Barberis, Shleifer and Vishny [5] modeled investor sentiment as conservatism plus representativeness, giving under- and over-reaction — the psychological microfoundation for winner-chasing flows.

Regime dependence of the drift follows Hamilton [6], whose Markov-switching model of nonstationary economic time series is the direct antecedent of the VIX-split transition pair $(\mathbf{T}_{\text{calm}}, \mathbf{T}_{\text{stress}})$ used here.

What is new here is the combination: a forward-time cluster-dynamics ODE framework ported from irradiated-materials modeling, with the market regime as the cluster, stocks as the migrating particles, a bilinear herding “reaction” as the tested concentration mechanism, and two out-of-sample hindcast episodes (2020–21, 2008–09) on measured S&P 500 data.

3 Model formulation

3.1 State, clusters and regimes

The state is $\mathbf{c}(t) \in \mathbb{R}^{15}$ with $c_k(t)$ the fraction of S&P 500 constituents in regime bin $k = 3m + v$, $m = 0, \dots, 4$ the momentum bin, $v = 0, 1, 2$ the volatility bin. Momentum for stock i at month-end t is the 12-minus-1 month total return on adjusted close (measured); volatility is the annualized standard deviation of daily log returns over the trailing 60 trading days (measured). Bins are absolute (Sec. 1), recomputed from the measured features each month; the panel runs 2000-12 through 2022-01 (253 months, 106,885 stock-months; the constituent count grows 343→486 over the window).

3.2 Master equation in matrix form

The dynamics take the compact form

$$\frac{d\mathbf{c}}{dt} = \mathbf{P}(t) + \mathbf{S}\mathbf{J}(\mathbf{c}) - \mathbf{D}(t)\mathbf{c}, \quad (1)$$

with the pieces identified as follows.

- **Source vector:** $\mathbf{P}(t) \in \mathbb{R}^{15}$ — index additions $s(t)$ enter uniformly across bins (measured mean 8×10^{-4} /month); removals are proportional at rate $d(t)$ (measured ≈ 0). Net effect on the distribution is negligible; the state is renormalized each step.
- **Stoichiometry and flux:** the drift is the linear part $\mathbf{T}^\top \mathbf{c}$ with $\mathbf{T}$ the measured monthly Markov transition matrix (discrete-time form, Sec. 3). The herding flux is the bilinear term

$$H_k(\mathbf{c}) = c_k \frac{q_k - \bar{q}(\mathbf{c})}{4}, \quad \bar{q}(\mathbf{c}) = \sum_j c_j q_j, \quad (2)$$

where $q_k \in \{0, \dots, 4\}$ is the momentum bin of cluster k . H is zero-sum by construction ($\sum_k H_k = 0$): it moves mass from below-average-momentum bins to above-average-momentum bins at rate h , the “rich get richer” ratchet. It is the market counterpart of the assortative-mating flux: bilinear in the state, sign-indefinite, vanishing when $h = 0$.

- **Regime-dependent drift:** in the extended model the drift operator switches with the volatility regime, $\mathbf{T}(t) = \mathbf{T}_{\text{stress}}$ when the monthly mean VIX exceeds 20 and $\mathbf{T}_{\text{calm}}$ otherwise,

both measured on pre-episode data (31,225 stress transitions for the 2000–2019 calibration, including 37 dot-com and 27 financial-crisis months).

In discrete monthly time the integrated form is

$$\mathbf{c}(t+1) = \mathbf{T}(t)^\top \mathbf{c}(t) + h \mathbf{H}(\mathbf{c}(t)) + \mathbf{s} - d \mathbf{c}(t), \quad (3)$$

renormalized to sum 1 each step.

3.3 Why these vertices and edges

Momentum and volatility are the two measured features along which equity crowding episodes visibly separate: in every historical concentration event the winner cohort is simultaneously high-momentum and elevated-volatility, while the loser cohort pools in the low-momentum bins. Absolute bins keep the cluster semantics of the materials framework (a cluster *size* is a fixed state, not a rank). Drift is measured rather than modeled because the month-to-month migration statistics are directly observable; herding is the single free mechanism parameter h because winner-chasing is the hypothesis under test, and isolating it as one bilinear term keeps the null ($h = 0$) sharp.

4 Market-dynamics graph

The reaction graph below is drawn from the model specification: every vertex is a real state variable c_k , every edge a real term of the right-hand side of Eq. (3). Vertices are the 15 regime clusters R_{mv} on the momentum (m) $\times$ volatility (v) lattice; blue edges are the dominant measured drift transitions ($T_{ij} > 0.08$); the red arrow is the herding ratchet, which carries mass rightward toward the winner column at rate h ; index entry/exit attach as source/sink.

5 Reaction kernel table

Every admissible transition (edge class) of the graph, with its stoichiometry and rate kernel — the direct analog of the RadCluster reaction table.

#	Process class	Transition	Kernel / flux
1	Index entry (source)	$\emptyset \rightarrow R_{mv}$ (uniform)	$s/15$ per bin, $s = 8 \times 10^{-4}/\text{mo}$ [measured]
2	Drift (Markov)	$R_i \rightarrow R_j$	$T_{ij} c_i$, $\mathbf{T}$ measured 2000–2019
3	Regime drift	$R_i \rightarrow R_j$ under $\mathbf{T}_{\text{calm}} / \mathbf{T}_{\text{stress}}$	VIX-split pair, both measured pre-episode
4	Herding (nonlinear)	mass $R_i \rightsquigarrow R_j$, $q_j > q_i$	$h H_k(\mathbf{c})$ of Eq. (2), zero-sum
5	Index exit (sink)	$R_{mv} \rightarrow \emptyset$	$d c_{mv}$, $d \approx 0$ [measured]

The herding kernel is bilinear in $\mathbf{c}$ (through c_k directly and through $\bar{q}(\mathbf{c})$), sign-indefinite, and the only term carrying the free mechanism parameter h . Drift is linear and fully measured; the regime pair ($\mathbf{T}_{\text{calm}}, \mathbf{T}_{\text{stress}}$) keeps the same form with VIX-conditioned operators.

6 Input parameters and assumptions

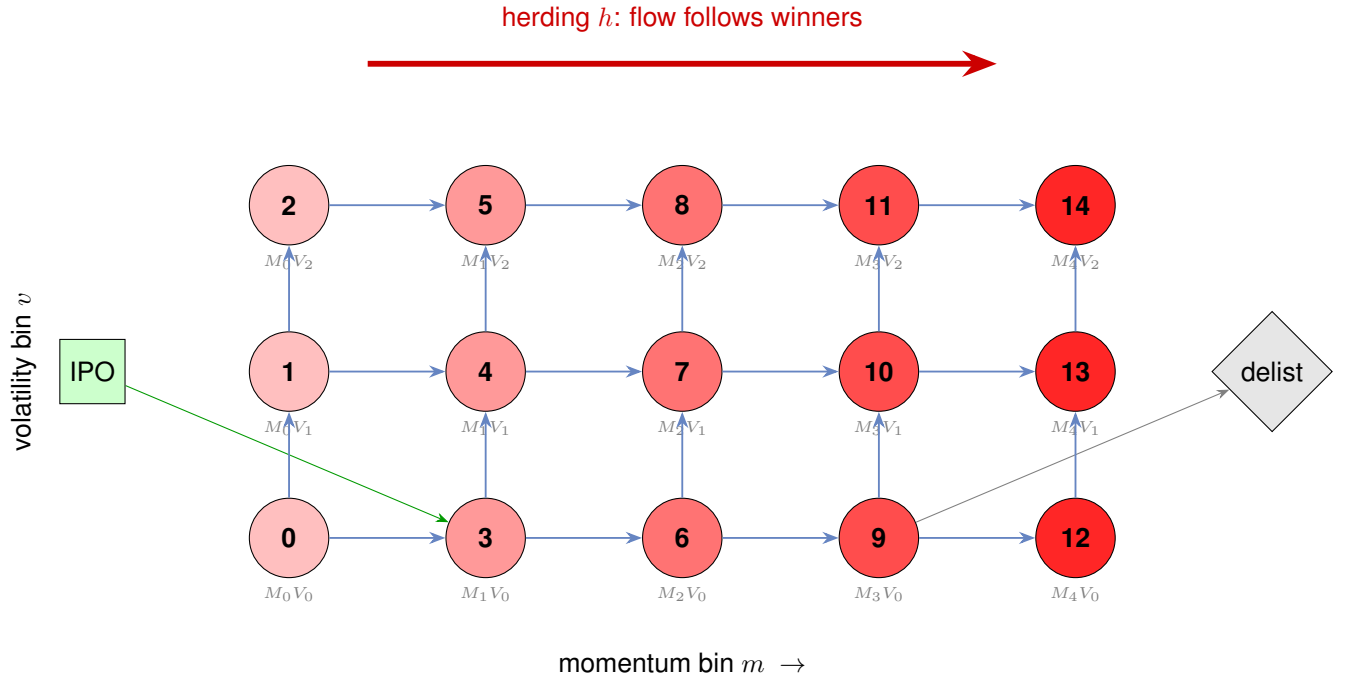


Figure 1: Reaction-admissibility graph of the market-regime model, drawn from the specification. Vertices: the 15 regime clusters R_{mv} (bin index and $M_m V_v$ label). Edges: measured drift transitions (blue), index entry/exit (green/gray), and the herding ratchet toward the winner column (red). Node shading deepens with momentum bin.

Parameter	Symbol	Value / series	Source	Quality
Prices	$p_i(t)$	daily adj. close, 486 tickers, 2000–2021	Yahoo Finance [7]	measured
Constituents	—	503 current S&P 500 members	s-and-p-500-companies [8]	measured
Momentum	$r_i(t)$	12-minus-1 month return on adj. close	computed from Yahoo prices [7]	measured
Volatility	$\sigma_i(t)$	trailing-60d ann. stdev of log returns	computed from Yahoo prices [7]	measured
VIX	$V(t)$	daily close, monthly mean; stress > 20	CBOE ^VIX via Yahoo [7]	measured
Drift	$\mathbf{T}$	15 × 15 monthly transitions, 2000–2019	counted from the regime panel	measured
Regime drift	$\mathbf{T}_{\text{calm}}, \mathbf{T}_{\text{stress}}$	63,592 calm / 31,225 stress transitions (2000–2019)	VIX-split counts from the regime panel	measured
Herding rate	h	0 (null); swept 0–3; best 0 calm, 0.2 stress	grid calibration on pre-episode data	calibrated
Index entry	s	8×10^{-4} /month, uniform	panel appearance rate	measured
Index exit	d	≈ 0	panel disappearance rate	measured
Bin edges	—	mom: < 0, 0–15, 15–30, 30–60, > 60%; vol: < 20, 20–32, > 32%	set from empirical quantiles	assumed
VIX threshold	V^*	20	separates historical stress months	assumed
Initial state	$\mathbf{c}(t_0)$	observed distribution at episode start	regime panel	measured
Survivorship	—	panel 343→486 stocks; delisted losers missing early	current-constituent backfill	assumed (bias noted)

Validation: two out-of-sample hindcasts (Sec. 6). The 2016–2019 calm calibration selects $h = 0$ with RMSE 0.0451 on bin shares, i.e. drift alone explains the tranquil distribution. Survivorship bias: backfilling current constituents understates early loser-bin mass (delisted stocks are disproportionately losers), so historical drawdown concentrations are lower bounds.

7 Results

Scenario structure. **Case 1 (2020–21 COVID episode):** hindcast January 2020→December 2021 from the observed January 2020 distribution, drift calibrated 2000–2019 — the featured case. The episode is a recovery-driven concentration: the March 2021 twelve-month winner share ($> 60\%$ bin) reaches 55% measured off the crash bottom. **Case 2 (2008–09 financial crisis):** hindcast January 2008→ December 2009 from the observed January 2008 distribution, drift calibrated 2000–2007. The episode is a drawdown: 95% of stocks sit in the loser bin ($< 0\%$) in late 2008.

Calm-period calibration. On 2016–2019 the herding sweep selects $h = 0$ decisively (RMSE 0.0451 at $h = 0$, rising monotonically to 0.1444 at $h = 3$): in tranquil markets the measured drift explains the regime distribution and winner-chasing adds nothing.

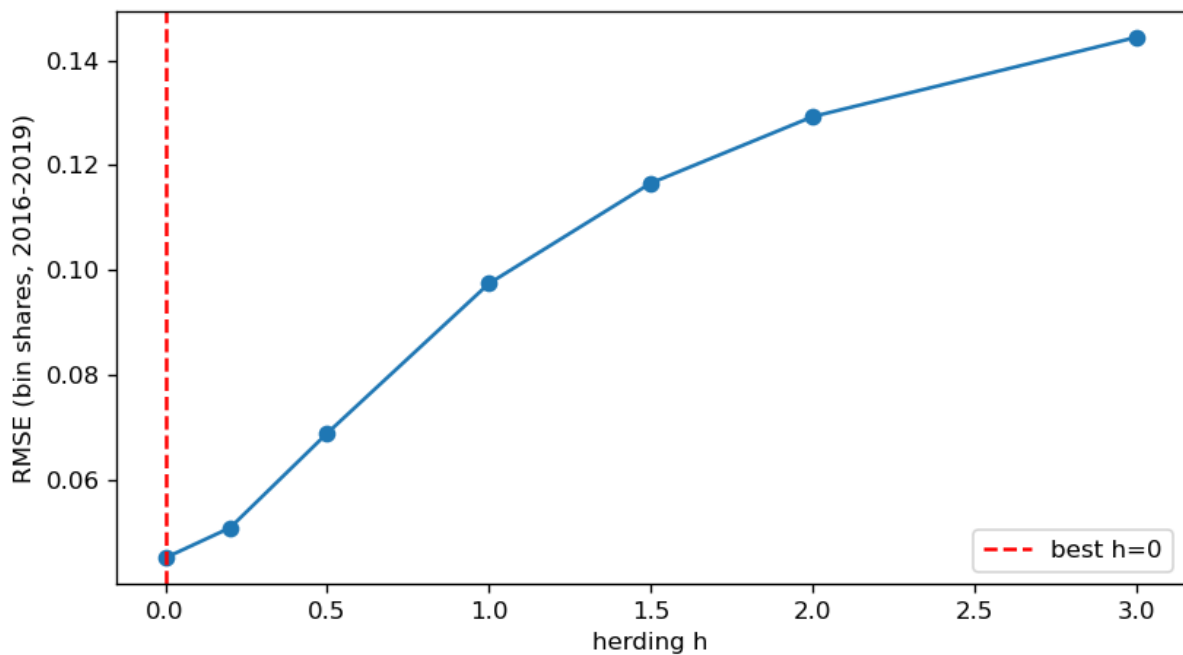


Figure 2: Herding calibration on 2016–2019: RMSE of predicted vs. observed bin shares vs. h . The null $h = 0$ wins; herding degrades the calm-period fit monotonically.

7.1 Case 1: the 2020–21 concentration

With the long-history calibration (2000–2019), the regime-switching model with $h = 0.2$ attains RMSE 0.0821, about 8% better than stationary drift (0.0888):

Model	RMSE (bin shares)
Stationary $\mathbf{T}$, $h = 0$	0.0888
Stationary $\mathbf{T}$, $h = 0.2$	0.0865
Regime ($\mathbf{T}_{\text{calm}}, \mathbf{T}_{\text{stress}}$), $h = 0$	0.0872
Regime + $h = 0.2$	0.0821

The regime operator bends toward the concentration while stationary drift stays flat (Fig. 3). Mechanism first: the earlier short-history failure (Sec. 1, REPORT.md) was data starvation — with only 5 stress months $\mathbf{T}_{\text{stress}}$ had a 0% stationary winner share and actively pushed mass away from winners. With 31,225 stress transitions its stationary top-bin share is 5.6%: the operator finally knows winner dynamics exist.

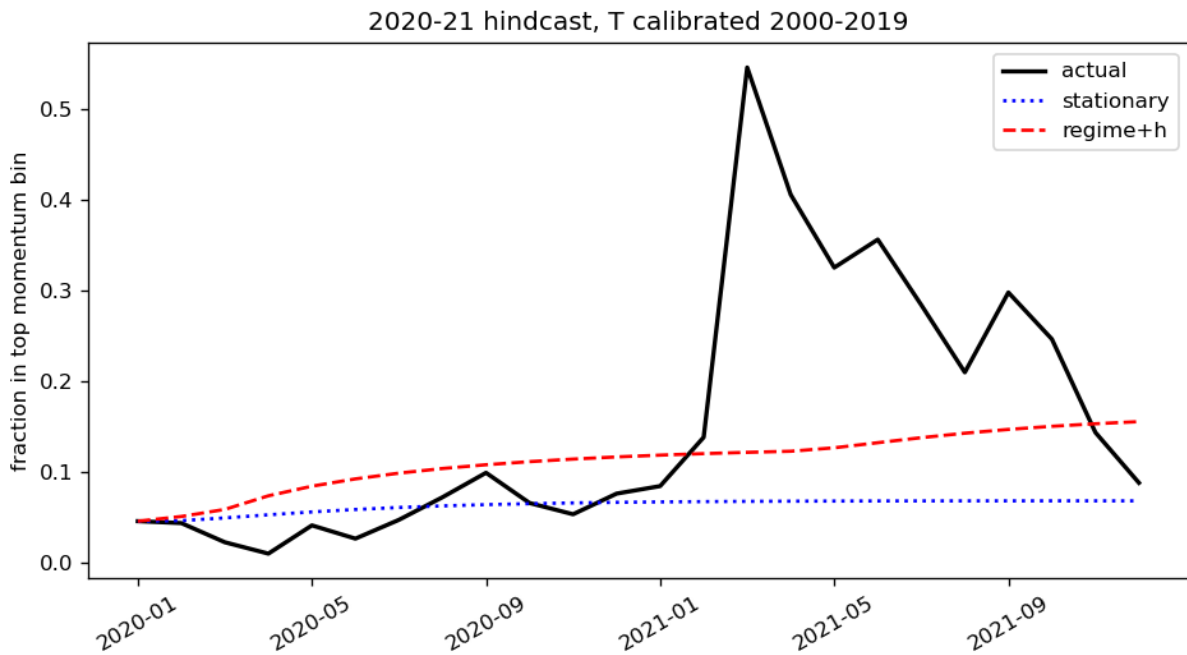


Figure 3: Case 1 hindcast: fraction of stocks in the top momentum bin ($> 60\%$ twelve-month return), actual vs. stationary drift vs. regime-switching drift with herding. Neither model captures the full March 2021 spike (55%), but the regime model rises toward the concentration while stationary drift stays flat.

7.2 Case 2: the 2008–09 drawdown

Calibrated 2000–2007 (11,538 stress transitions), the regime model wins outright at $h = 0$ (RMSE 0.1110 vs. 0.1293 stationary, $\sim 14\%$ improvement) — and herding *hurts* (0.1431):

Model	RMSE (bin shares)
Stationary $\mathbf{T}$, $h = 0$	0.1293
Stationary $\mathbf{T}$, $h = 0.2$	0.1431
Regime ($\mathbf{T}_{\text{calm}}, \mathbf{T}_{\text{stress}}$), $h = 0$	0.1110
Regime + $h = 0.2$	0.1269

The herding flux pushes mass toward winners while crash dynamics drain into losers, so $h > 0$ is actively wrong in a drawdown. The mechanism conclusion: h is not a universal constant — its useful sign depends on episode type (recovery vs. crash). Regime $\mathbf{T}$ at $h = 0$ is the most robust single model: it wins Case 2 outright and is second in Case 1.

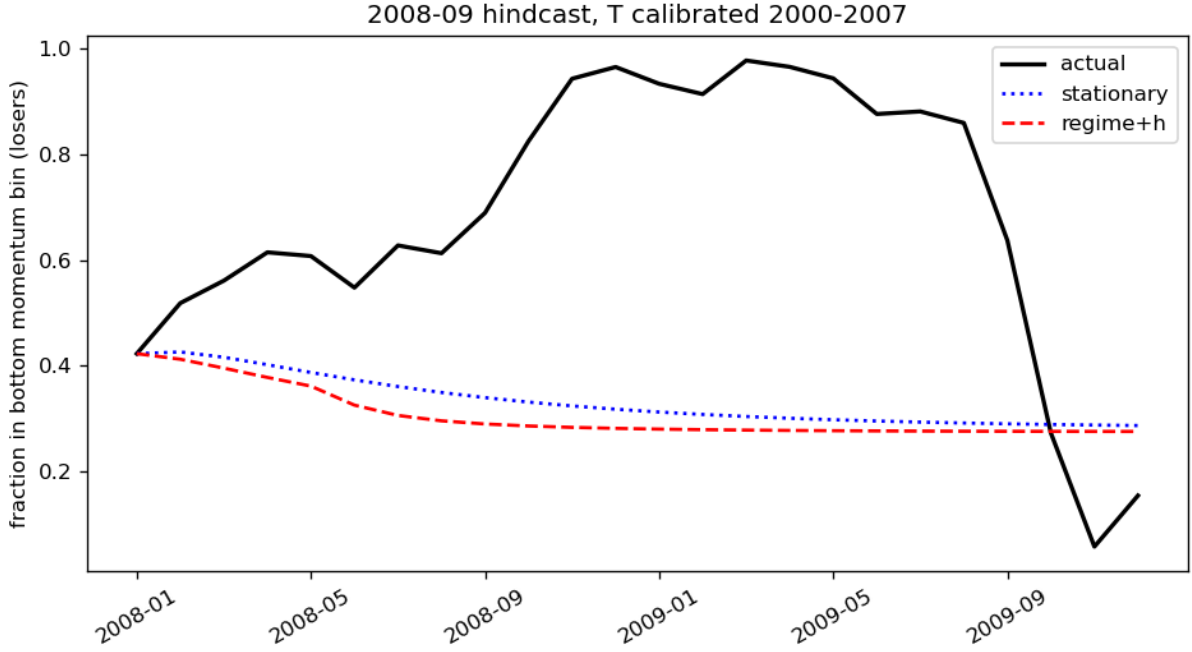


Figure 4: Case 2 hindcast: fraction of stocks in the bottom momentum bin (losers, $< 0\%$), actual vs. stationary drift vs. regime-switching drift with herding. The drawdown drives 95% of stocks into the loser bin by late 2008; the regime model tracks the early phase better. (Survivorship: delisted losers are missing, so the observed concentration is a lower bound.)

Distribution snapshots. Figure 5 shows observed vs. predicted regime distributions at three dates: the COVID shock (June 2020, mass piled in the loser column), the vaccine-recovery (December 2021), and the featured concentration shape.

Amplitude gap. Neither case’s full spike is captured (55% top-bin in March 2021; 95% bottom-bin in late 2008). The monthly Markov drift mean-reverts too fast toward its stationary distribution; the additive herding flux of Eq. (2) can only redistribute mass the drift already moves. This is the honest boundary of the prototype and the first item of Sec. 7.

8 Suggestions for more accurate models and data

1. **Multiplicative herding on rates.** Let h modulate the transition rates themselves, $T_{ij}(h) \propto T_{ij} \exp(h q_j)$, rather than adding an independent flux. An amplifier acting on existing flows can generate sharper concentrations than the additive perturbation of Eq. (2).
2. **Episode-signed herding.** Case 1 vs. Case 2 show the useful sign of h flips between recovery and crash. Condition h on the VIX *change* (rising vs. falling), not just its level, or fit $h_{\text{crash}} < 0 < h_{\text{recovery}}$.
3. **Memory in the drift.** A first-order Markov operator forgets too fast. Second-order transitions or a slow-moving latent factor would let concentrations persist at observed amplitudes.
4. **Capitalization weighting.** The state is currently equal-weight (stock counts). The 2020–21 narrative is cap-weighted (mega-cap tech); historical shares-outstanding would let $c(t)$ track capital rather than headcount.

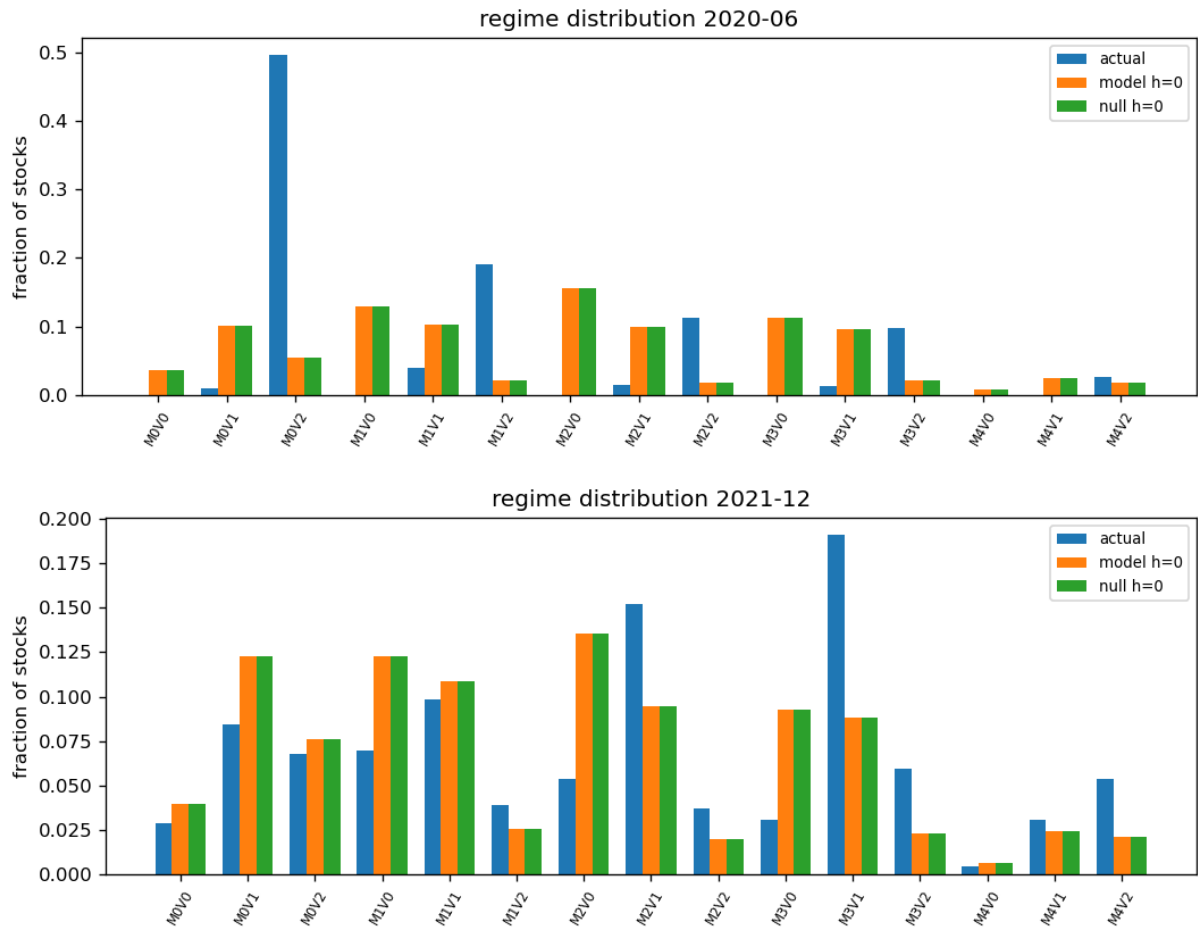


Figure 5: Regime-distribution snapshots: observed vs. model (h best) vs. drift-only null, June 2020 (top) and December 2021 (bottom). Bin labels $M_m V_v$: momentum bin m , volatility bin v .

5. **Historical membership.** Replace the backfilled current constituents with true historical S&P 500 membership (CRSP) to remove the survivorship bias, which currently understates drawdown concentrations.
6. **Finer volatility resolution and intraday features.** Three volatility bins are coarse; turnover/liquidity as a third feature axis is the natural next vertex dimension.
7. **Out-of-sample rolling forecasts.** Replace the two-episode hindcast with rolling 12-month-ahead forecasts over the full window, reporting probabilistic skill (CRPS on bin shares) rather than RMSE alone.

9 References

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